



Title

rdplot — Data-Driven Regression Discontinuity Plots.

Syntax

```
rdplot depvar indepvar [if] [in] [, c(#) nbins(# #) binselect(binmethod) scale(# #) support(# #) p(#) h(# #) kernel(kernelfn) weights(weightsvar) covs(covars) covs_eval(covars_eval) covs_drop(covsdropoption) masspoints(masspointsoption) precision(precopt) ci(cilevel) shade graph_options(gphopts) hide genvars ]
```

Description

rdplot implements several data-driven Regression Discontinuity (RD) plots, using either evenly-spaced or quantile-spaced partitioning. Two type of RD plots are constructed: (i) RD plots with binned sample means tracing out the underlying regression function, and (ii) RD plots with binned sample means mimicking the underlying variability of the data. For technical and methodological details see [Calonico, Cattaneo and Titiunik \(2015a\)](#).

Companion commands are: **rdrobust** for point estimation and inference procedures, and **rdbwselect** for data-driven bandwidth selection.

A detailed introduction to this command is given in [Calonico, Cattaneo and Titiunik \(2014\)](#), and [Calonico, Cattaneo, Farrell and Titiunik \(2017\)](#). A companion R package is also described in [Calonico, Cattaneo and Titiunik \(2015b\)](#).

Related Stata and R packages useful for inference in RD designs are described in the following website:

<https://rdpackages.github.io>

Requires Stata 16 or later.

Options

Estimand

c(#) specifies the RD cutoff in *indepvar*. Default is **c**(0).

Bin Selection

nbins(# #) specifies the number of bins used to the left of the cutoff, denoted J_- , and to the right of the cutoff, denoted J_+ , respectively. If not specified, J_+ and J_- are estimated using the method and options chosen below.

binselect(*binmethod*) specifies the data-driven procedure to select the number of bins. This option is available only if J_- and J_+ are not set manually using **nbins**(.). Options are:

es IMSE-optimal evenly-spaced method using spacings estimators.

espr IMSE-optimal evenly-spaced method using polynomial regression.

esmv mimicking variance evenly-spaced method using spacings estimators.

esmvpr mimicking variance evenly-spaced method using polynomial regression.

qs IMSE-optimal quantile-spaced method using spacings estimators.

qspr IMSE-optimal quantile-spaced method using polynomial regression.

qsmv mimicking variance quantile-spaced method using spacings estimators.

qsmvpr mimicking variance quantile-spaced method using polynomial regression.

Default is **binselect**(**esmv**).

Note: procedures involving spacing estimators are not invariant to rearrangements of *depvar* when there are repeated values (i.e., mass points in the running variable).

scale(# #) specifies multiplicative factors, denoted s_- and s_+ , respectively, to adjust the number of bins selected. Specifically, the number of bins used for the treatment and control groups will be $\text{ceil}(s_- * J_-)$ and $\text{ceil}(s_+ * J_+)$, where J_- and J_+ denote the optimal numbers of bins originally computed for each group. Default is **scale(1 1)**.

support(# #) sets an optional extended support of the running variable to be used in the construction of the bins. Default is the sample range.

masspoints(*masspointsoption*) checks and controls for repeated observations in the running variable. Options are:
off ignores the presence of mass points.
check looks for and reports the number of unique observations at each side of the cutoff.
adjust sets **binselect**(*binmethod*) as polynomial regression when mass points are present.
 Default option is **masspoints(adjust)**.

precision(*precopt*) controls the storage precision of internal temporary variables (the **rdplot_*** output variables when **genvars** is requested). Options are **double** (default) and **single**; **single** maps to Stata's **float** storage type. Default is **precision(double)**.

Polynomial Fit

p(#) specifies the order of the (global) polynomial fit used to approximate the population conditional expectation functions for control and treated units. Default is **p(4)**.

h(# #) specifies the bandwidth used to construct the (global) polynomial fits given the kernel choice **kernel(.)**. If not specified, the bandwidths are chosen to span the full support of the data. If two bandwidths are specified, the first bandwidth is used for the data below the cutoff and the second bandwidth is used for the data above the cutoff.

kernel(*kernelfn*) specifies the kernel function used to construct the local-polynomial estimator(s). Options are: **triangular**, **epanechnikov**, and **uniform**. Default is **kernel(uniform)** (i.e., equal/no weighting to all observations on the support of the kernel).

weights(*weightsvar*) is the variable used for optional weighting of the estimation procedure. The unit-specific weights multiply the kernel function.

covs(*covars*) additional covariates used to construct the local-polynomial estimator(s).

covs_eval(*covars_eval*) sets the evaluation points for the additional covariates, when included in the estimation. Options are: **0** and **mean** (default).

covs_drop(*covsdroptoption*) assess collinearity in additional covariates used for estimation and inference. Options **pinv** (default choice) and **invsym** drops collinear additional covariates, differing only in the type of inverse function used. Option **off** omits the check for collinear additional covariates.

Plot Options

ci(*cilevel*) graphical option to display confidence intervals of level *cilevel* for each bin.

shade graphical option to replace confidence intervals with shaded areas.

graph_options(*gphopts*) graphical options to be passed on to the underlying graph command.

hide omits the RD plot.

Generate Variables

genvars generates new variables storing the following results.

- rdplot_id** unique bin ID for each observation. Negative natural numbers are assigned to observations to the left of the cutoff, and positive natural numbers are assigned to observations to the right of the cutoff.
- rdplot_N** number of observations in the corresponding bin for each observation.
- rdplot_min_bin** lower end value of the bin for each observation.
- rdplot_max_bin** upper end value of the bin for each observation.
- rdplot_mean_bin** middle point of the corresponding bin for each observation.
- rdplot_mean_x** sample mean of the running variable within the corresponding bin for each observation.
- rdplot_mean_y** sample mean of the outcome variable within the corresponding bin for each observation.
- rdplot_se_y** standard deviation of the mean of the outcome variable within the corresponding bin for each observation.
- rdplot_ci_l** lower end value of the confidence interval for the sample mean of the outcome variable within the corresponding bin for each observation.
- rdplot_ci_r** upper end value of the confidence interval for the sample mean of the outcome variable within the corresponding bin for each observation.
- rdplot_hat_y** predicted value of the outcome variable given by the global polynomial estimator.

Example: Cattaneo, Frandsen and Titiunik (2015) Incumbency Data

```

Setup
. use rdrobust_senate.dta

Basic specification with title
. rdplot vote margin, graph_options(title(RD Plot))

Quadratic global polynomial with confidence bands
. rdplot vote margin, p(2) ci(95) shade

```

Stored results

rdplot stores the following in **e()**:

Scalars

e(N)	number of observations
e(N_l)	original number of observations to the left of the cutoff
e(N_r)	original number of observations to the right of the cutoff
e(N_h_l)	effective number of observations (given the bandwidth) to the left of the cutoff
e(N_h_r)	effective number of observations (given the bandwidth) to the right of the cutoff
e(c)	cutoff value
e(p)	order of the polynomial used for the fit
e(J_star_l)	selected number of bins to the left of the cutoff
e(J_star_r)	selected number of bins to the right of the cutoff

Macros

e(cmd)	rdplot
e(cmdline)	command as typed
e(title)	title (RD plot)
e(depvar)	name of dependent (outcome) variable
e(runningvar)	name of running variable
e(binselect)	method used to compute the optimal number of bins
e(eq_l)	polynomial equation (left of cutoff) as a string, suitable for overlaying the fit via twoway function
e(eq_r)	polynomial equation (right of cutoff) as a string, suitable for overlaying the fit via twoway function
e(precision)	storage precision selected via precision(): double (default) or single

Matrices

e(coef_l) coefficients of the p -th order polynomial estimated to the left of the cutoff
e(coef_r) coefficients of the p -th order polynomial estimated to the right of the cutoff
e(coef_covs) coefficients of the additional covariates, only returned when **covs()** are used

References

- Calonico, S., M. D. Cattaneo, M. H. Farrell, and R. Titiunik. 2017. rdrobust: Software for Regression Discontinuity Designs. *Stata Journal* 17(2): 372-404.
- Calonico, S., M. D. Cattaneo, and R. Titiunik. 2014b. Robust Data-Driven Inference in the Regression-Discontinuity Design. *Stata Journal* 14(4): 909-946.
- Calonico, S., M. D. Cattaneo, and R. Titiunik. 2015a. Optimal Data-Driven Regression Discontinuity Plots. *Journal of the American Statistical Association* 110(512): 1753-1769.
- Calonico, S., M. D. Cattaneo, and R. Titiunik. 2015b. rdrobust: An R Package for Robust Nonparametric Inference in Regression-Discontinuity Designs. *R Journal* 7(1): 38-51.
- Cattaneo, M. D., B. Frandsen, and R. Titiunik. 2015. Randomization Inference in the Regression Discontinuity Design: An Application to Party Advantages in the U.S. Senate. *Journal of Causal Inference* 3(1): 1-24.

Authors

Sebastian Calonico, University of California, Davis, CA. scalonico@ucdavis.edu.

Matias D. Cattaneo, Princeton University, Princeton, NJ. matias.d.cattaneo@gmail.com.

Max H. Farrell, University of California, Santa Barbara, CA. mhfarrell@gmail.com.

Rocio Titiunik, Princeton University, Princeton, NJ. rocio.titiunik@gmail.com.